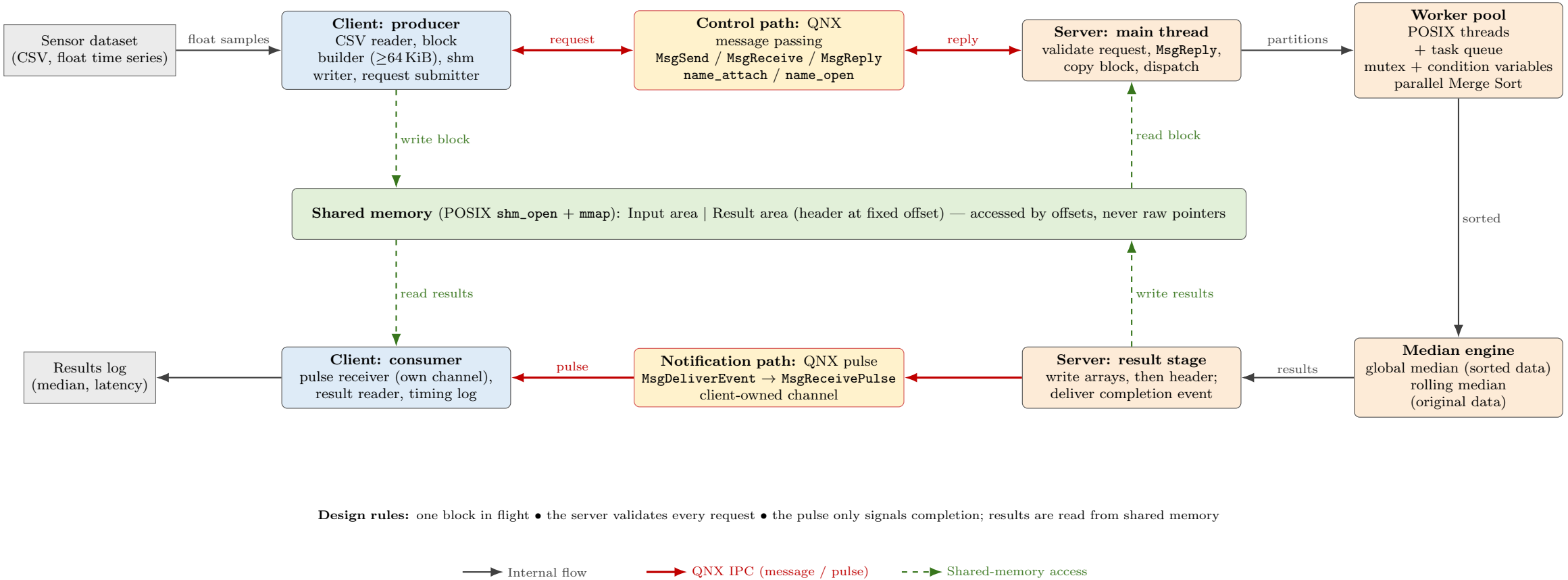

Sensor dataset (CSV, float time series)
float samples
Client: producer
CSV reader, block builder (≥64 KiB), shm writer, request submitter
request
Control path: QNX
message passing
MsgSend / MsgReceive / MsgReply
name_attach / name_open
reply
Server: main thread
validate request, MsgReply, copy block, dispatch
partitions
Worker pool
POSIX threads
+ task queue
mutex + condition variables
parallel Merge Sort
write block
read block
Shared memory (POSIX shm_open + mmap): Input area | Result area (header at fixed offset) — accessed by offsets, never raw pointers
sorted
read results
write results
Results log (median, latency)
Client: consumer
pulse receiver (own channel), result reader, timing log
pulse
Notification path: QNX pulse
MsgDeliverEvent → MsgReceivePulse
client-owned channel
Server: result stage
write arrays, then header; deliver completion event
results
Median engine
global median (sorted data)
rolling median
(original data)
Design rules: one block in flight • the server validates every request • the pulse only signals completion; results are read from shared memory
Internal flow
QNX IPC (message / pulse)
Shared-memory access